Supplementary Material

A pre-registered evaluation of language-model agents for primary transcription-factor identification

`DeconfTF-Bench` Protocol & Audit Suite

November 2026

Overview of Supplementary Files

This Supplementary PDF contains the complete printed versions of **Supplementary Table S1** (51 QC-passing non-KnockTF GEO bulk RNA-seq contrasts), **Supplementary Table S2** (pre-registration deviations D1–D8), and **Supplementary Table S5** (R3 three-dimensional benchmark audit across 11 deterministic operators). The accompanying multi-sheet workbook `Supplementary_Tables_S1_to_S5.xlsx` provides machine-readable spreadsheets for all tables, including **Supplementary Table S3** (514 Replogle K562 and 78 RPE1 Perturb-seq profiles), **Supplementary Table S4** (GEO title-level, GSM-contrast, and Phase-1 QC screening logs), and per-experiment E5 agent predictions across GEO ($N = 51$), KnockTF Unseen ($N = 64$), and Replogle K562 ($N = 267$).

Supplementary Table S1: 51 Pre-Registered Non-KnockTF GEO Contrasts

Contrast ID	GSE	Target TF	Cell Line	Genes	Target $\log_2$FC	ULM	M5	M9	M10
GSE263639_BLaER_CTCF_Degron	GSE263639	CTCF	BLaER	29923	0.14	353	71	38	48
GSE267370_Jurkat_MYB_Degron	GSE267370	MYB	Jurkat	18850	0.19	318	25	16	20
GSE289670_Molm13_TAF1_Degron	GSE289670	TAF1	Molm13	19202	-0.40	244	115	182	180
GSE114742_MCF10AT_EP300_KD	GSE114742	EP300	MCF10AT	15021	-0.73	173	92	149	152
GSE179314_HCT116_SP3_KD	GSE179314	SP3	HCT116	19985	-1.14	14	21	21	28
GSE202868_MDAMB231_JARID2_KD	GSE202868	JARID2	MDA-MB-231	24154	-1.27	598	431	431	424
GSE205659_528NS_SREBF1_KD	GSE205659	SREBF1	528NS	18918	-1.11	631	41	26	29
GSE205659_83NS_SREBF1_KD	GSE205659	SREBF1	83NS	18918	-1.46	63	1	1	1
GSE212684_MGC803_ATF2_KD	GSE212684	ATF2	MGC803	29885	-1.29	630	156	203	138
GSE227040_KGN_FOXL2_KD	GSE227040	FOXL2	KGN	19211	-1.80	339	617	619	637
GSE239779_HeLa_ZBTB17_KD	GSE239779	ZBTB17	HeLa	19205	-0.90	81	53	151	137
GSE260921_DU145_NFIB_KD	GSE260921	NFIB	DU145	15624	-0.85	118	469	480	466
GSE264653_U2OS_MITF_KD	GSE264653	MITF	U2OS	29786	-3.20	114	308	308	96
GSE275694_FT190_SMARCA4_KD	GSE275694	SMARCA4	FT190	41006	-2.78	303	427	427	431
GSE279748_BeWo_GCM1_KD	GSE279748	GCM1	BeWo	19100	-0.72	4	25	140	108
GSE279807_Myla_FOXM1_KD	GSE279807	FOXM1	Myla	48767	-1.74	25	4	117	5
GSE290232_TMD8_IRF4_KD	GSE290232	IRF4	TMD8	24369	-1.20	638	13	13	9
GSE293321_HTR8_MYBL2_KD	GSE293321	MYBL2	HTR-8_SVneo	19232	-1.84	454	123	194	204
GSE293321_hTSC_MYBL2_KD	GSE293321	MYBL2	hTSC	19235	-2.35	308	20	127	145
GSE293430_OVCA433_NR1H3_KD	GSE293430	NR1H3	OVCA433	19210	-0.94	603	543	548	317
GSE296237_LNCaP_HOXB13_KD	GSE296237	HOXB13	LNCaP	26418	-2.35	539	251	88	118
GSE299180_hESC_CTCF_KD	GSE299180	CTCF	primary_endometr	19028	-1.44	548	276	95	122
GSE299679_H146_KLF9_KD	GSE299679	KLF9	NCI-H146	17164	-1.15	642	158	68	87
GSE300616_HemEC_EPAS1_KD	GSE300616	EPAS1	HemECs	19235	-2.73	623	433	442	553
GSE309470_MDAMB231_ENO1_KD	GSE309470	ENO1	MDA-MB-231	19233	-4.84	159	400	416	414
GSE312184_HepG2_EZH2_KD	GSE312184	EZH2	HepG2	26278	-0.93	247	118	194	216
GSE316581_MCF7_HOXB2_KD	GSE316581	HOXB2	MCF-7	19236	-1.15	392	449	449	454
GSE330070_HUVEC_SMAD4_KD	GSE330070	SMAD4	HUVEC	18965	-1.72	235	77	39	50
GSE337367_HUDEP2_EZH2_KD	GSE337367	EZH2	HUDEP-2	48781	-1.56	218	268	300	306
GSE339268_HUCCT1_PBX1_KD	GSE339268	PBX1	HUCCT1	48781	-1.77	569	465	479	483
GSE339268_RBE_PBX1_KD	GSE339268	PBX1	RBE	48781	-2.57	512	522	530	530
GSE344516_NPC_YBX1_KD	GSE344516	YBX1	Primary_nucleus_	40991	-4.24	507	486	486	187
GSE175944_hiPSC_PITX2_KO	GSE175944	PITX2	hiPSC_aEHT	47821	-0.08	575	101	175	213
GSE229351_hESC_BACH1_KO	GSE229351	BACH1	hESC_VSMC	19211	0.25	87	115	115	61
GSE231949_CD34_RUNX1_KO	GSE231949	RUNX1	CD34_HSPC	18061	-0.08	14	11	11	11
GSE231960_RPMI8402_MYC_KO	GSE231960	MYC	RPMI-8402	18867	0.14	26	1	1	1
GSE245497_H9_HSF1_KO	GSE245497	HSF1	H9_hPSC_cardiomy	50164	-0.77	632	617	619	518
GSE246363_MCF7_FOXA1_KO	GSE246363	FOXA1	MCF-7	19162	-0.58	40	92	92	116
GSE252761_CVB_GTF2I_KO	GSE252761	GTF2I	CVB_cortical_org	25298	-0.18	453	475	487	485
GSE256043_WCM154_EZH2_KO	GSE256043	EZH2	WCM154	27960	0.37	510	269	318	336
GSE270149_RPE_EWSR1_KO	GSE270149	EWSR1	RPE_p53null	40861	-2.36	465	303	335	354
GSE272398_KOPN8_ZNF217_KO	GSE272398	ZNF217	KOPN-8	19215	0.21	137	613	615	629
GSE282831_JEG3_WWTR1_KO	GSE282831	WWTR1	JEG-3	50222	-3.03	377	395	416	411
GSE283179_2102Ep_TP53_KO	GSE283179	TP53	2102Ep	18867	-1.60	49	2	2	2
GSE283179_H9_TP53_KO	GSE283179	TP53	H9	18867	-1.54	53	1	1	1
GSE283179_LUMC007_TP53_KO	GSE283179	TP53	LUMC007	18867	-2.16	25	1	1	1
GSE283179_NCCIT_TP53_KO	GSE283179	TP53	NCCIT	18867	-0.70	71	9	8	8
GSE286216_HCT116_ARID1A_KO	GSE286216	ARID1A	HCT116	49625	-0.64	59	152	202	222
GSE295225_SCC25_RARG_KO	GSE295225	RARG	SCC-25	30476	-3.28	49	110	182	592

Contrast ID	GSE	Target TF	Cell Line	Genes	Target $\log_2$FC	ULM	M5	M9	M10
GSE306030_HaCaT_BCL11A_KO	GSE306030	BCL11A	HaCaT	19825	-0.14	215	314	345	160
GSE319443_A375_SOX10_KO	GSE319443	SOX10	A375	19188	-2.30	14	4	1	3

Supplementary Table S2: Pre-Registration Deviations (D1–D8)

ID	Item	Pre-registration rule	What changed & why	Impact / equivalence check	Changes conclusion?
D1	R3 audit script bug fix	External scoring calls frozen `deconftf_tools.py` unchanged	`run_r3_benchmark_audit.py` had `fillna(0)` on `est_d` before Z-score and lowercase `C11orf95` mask	After fix, 0/266 Dev and 0/64 Unseen rank differences vs committed KnockTF tables; frozen scorer untouched	No
D2	M10 soft-gate refreeze (R6)	R5 checksum of `develop_m10_soft_gate.py`	Expanded λ_{bind} grid on Dev 266 before GEO download; 168/170 LOSO folds selected $(1.0, 0.3)$	New SHA-256 `1e513f52...`; Unseen and Replogle were not used to pick parameters	No
D3	GEO contrast notes (R6.1)	Locked GSM contrast table	Added notes for 3 single-replicate-arm contrasts and `GSE290232` TMD14/TMD8 typo before expression download	GSM membership and INCLUDE/EXCLUDE flags unchanged (`73128cca...`)	No
D4	GEO parser repairs (R6.2)	Phase-1 QC before one-shot Phase-2 scoring	Two-pass HGNC symbol mapping, column readers, and SUM duplicate collapse after 37/49 engineering parse failures	QC pass rose from 25 to 51 contrasts; only pass counts were inspected before any rank scoring	No
D5	Duplicate symbol SUM vs average	Average duplicate HGNC symbols before $\log_2$ FC	Primary R6.2 used SUM for transcript-to-gene count aggregation; average run as sensitivity	Same 51 PASS contrasts; five-test Holm family still shows only M5 vs ULM significant	No
D6	M9 LOSO script recovery	LOSO over 3×3 grid chose $(0.5, 3.0)$ in 165/170 folds (27/266)	Original script unrecoverable; recomputation shows all folds tied on train Top-1 (26/266 by grid order; 27/266 preferring $(0.5, 3.0)$)	Frozen operator $(0.5, 3.0)$ unchanged; manuscript reports both tie-break rules	No
D7	Replogle missing-column assertion (E5)	v3 `build_prompt` required target TF column present and equal to 0	Allowed target TF column either absent from `mat_blind` (98/267 TFs in 8,248-gene panel) or present and equal to 0	Discovered before any Replogle call or E5 endpoint; 856 sent GEO/Unseen prompts replayed with 0 byte differences	No
D8	Divergent-transcript redaction (E5)	Prompt matches to target TF symbol replaced with `[PERT]` and counted	Redacted hyphenated DE30 gene tokens matching target TF (e.g. `CTBP1-DT`) to `[PERT]` while retaining $\log_2$ FC	Discovered before any Replogle call; 0 byte differences on 856 GEO/Unseen prompts; affected 2/534 Replogle v3 prompts	No

Supplementary Table S5: R3 Benchmark Audit Across 11 Methods (Strict Blinding)

Cohort (N)	Method	Top-1 Hits (%)	Top-5 Hits (%)	Top-10%	Median Rank	Mean $\log_2$ Rank	Poole
KnockTF Dev (266)	M1_ULM (CollecTRI)	15 (5.64%)	29 (10.90%)	15.41%	155.0	6.450	
KnockTF Dev (266)	M2_MLM (decoupleR)	13 (4.89%)	25 (9.40%)	12.78%	221.5	6.829	
KnockTF Dev (266)	M_WSUM (decoupleR)	15 (5.64%)	34 (12.78%)	19.55%	118.0	6.238	
KnockTF Dev (266)	M_WMEAN (decoupleR)	6 (2.26%)	10 (3.76%)	4.89%	203.5	7.326	
KnockTF Dev (266)	M_ZSCORE (decoupleR)	14 (5.26%)	26 (9.77%)	13.91%	164.0	6.556	
KnockTF Dev (266)	M_VIPER_Native (aREA)	13 (4.89%)	27 (10.15%)	11.65%	183.5	6.793	
KnockTF Dev (266)	M2b_VIPER_Shadow	16 (6.02%)	27 (10.15%)	13.91%	176.0	6.650	
KnockTF Dev (266)	M3_ChEA3_MeanRank	3 (1.13%)	9 (3.38%)	5.26%	169.0	6.856	
KnockTF Dev (266)	M4_ChEA3_TopRank	5 (1.88%)	17 (6.39%)	12.03%	150.0	6.513	
KnockTF Dev (266)	M5_MultiModal_Static	24 (9.02%)	47 (17.67%)	24.81%	68.0	5.618	
KnockTF Dev (266)	M9_Selective_Deconf	31 (11.65%)	56 (21.05%)	24.44%	80.5	5.543	
KnockTF Unseen (64)	M1_ULM (CollecTRI)	3 (4.69%)	6 (9.38%)	14.06%	169.0	6.486	
KnockTF Unseen (64)	M2_MLM (decoupleR)	4 (6.25%)	12 (18.75%)	21.88%	193.5	6.205	
KnockTF Unseen (64)	M_WSUM (decoupleR)	4 (6.25%)	6 (9.38%)	18.75%	130.5	6.324	
KnockTF Unseen (64)	M_WMEAN (decoupleR)	1 (1.56%)	1 (1.56%)	1.56%	215.5	7.343	
KnockTF Unseen (64)	M_ZSCORE (decoupleR)	3 (4.69%)	6 (9.38%)	15.62%	177.5	6.494	
KnockTF Unseen (64)	M_VIPER_Native (aREA)	2 (3.12%)	6 (9.38%)	14.06%	179.0	6.678	
KnockTF Unseen (64)	M2b_VIPER_Shadow	4 (6.25%)	9 (14.06%)	18.75%	171.0	6.427	
KnockTF Unseen (64)	M3_ChEA3_MeanRank	0 (0.00%)	1 (1.56%)	6.25%	92.5	6.382	
KnockTF Unseen (64)	M4_ChEA3_TopRank	1 (1.56%)	3 (4.69%)	14.06%	88.5	6.216	
KnockTF Unseen (64)	M5_MultiModal_Static	5 (7.81%)	9 (14.06%)	26.56%	42.0	5.281	
KnockTF Unseen (64)	M9_Selective_Deconf	8 (12.50%)	15 (23.44%)	26.56%	38.5	4.971	
Replogle RPE1 (76)	M1_ULM (CollecTRI)	2 (2.63%)	3 (3.95%)	5.26%	310.5	7.649	
Replogle RPE1 (76)	M_VIPER_Native (aREA)	1 (1.32%)	3 (3.95%)	5.26%	307.0	7.697	
Replogle RPE1 (76)	M4_ChEA3_TopRank	1 (1.32%)	1 (1.32%)	5.26%	368.5	7.902	
Replogle RPE1 (76)	M5_MultiModal_Static	3 (3.95%)	5 (6.58%)	9.21%	217.5	7.241	
Replogle RPE1 (76)	M9_Selective_Deconf	2 (2.63%)	5 (6.58%)	9.21%	261.5	7.475	